

Initial Sign Vectors in c_0 Refute a Schur-Generation Question

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June 16, 2026

Source Question

The source is Ondrej Kurka, *Tsirelson-like spaces and complexity of classes of Banach spaces*, arXiv:1612.09334. Remark 5.9(iii), on page 21, asks:

Does a Banach space X has necessarily the Schur property if it has a subset W such that $\overline{\text{co}}W = B_X$ and every weakly convergent sequence of elements of W is convergent in the norm?

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Remark 5.9. (i) It is possible to use [2, Theorem 3.2] to quantify the Schur property of $E_{\mathcal{T}}^*$ in a bit different direction. If the assumption $\forall \sigma \in 2^{\mathbb{N}} : \mathcal{T}(\sigma) \in \text{IF}$ is met, then $E_{\mathcal{T}}^*$ has the strong Schur property in the sense that every bounded sequence $a_1^*, a_2^*, \dots$ in $E_{\mathcal{T}}^*$ with $\inf\{\|a_n^* - a_m^*\|_{\mathcal{T}} : n \neq m\} > \varepsilon$ has a subsequence $a_{n_j}^*$ such that

$$\left\| \sum_{j=1}^k \lambda_j a_{n_j}^* \right\|_{\mathcal{T}} \geq \frac{\varepsilon}{12} \sum_{j=1}^k |\lambda_j|$$

for all $k \in \mathbb{N}$ and $\lambda_1, \dots, \lambda_k \in \mathbb{R}$.

(ii) If the assumption $\forall \sigma \in 2^{\mathbb{N}} : \mathcal{T}(\sigma) \in \text{IF}$ is met, a quantitative version of the Dunford-Pettis property of $E_{\mathcal{T}}^*$ and of $E_{\mathcal{T}}$ can be obtained as well (see [14, Proposition 6.4 and Theorem 6.5]).

(iii) In [20], a question was considered whether the proposed tree space method can be used for amalgamating of spaces with the Schur property (see [20, Remark 3.7(c)] for the exact formulation). Proposition 5.7(1) shows a concrete example of a family of spaces with the Schur property for which the answer is positive, although not trivial. Let us note that to prove that $E_{\mathcal{T}}^*$ has the Schur property would be much simpler if we had the positive answer to the following question: *Does a Banach space X has necessarily the Schur property if it has a subset W such that $\overline{\text{co}}W = B_X$ and every weakly convergent sequence of elements of W is convergent in the norm?*

Answer

The answer is negative. The counterexample is $X = c_0$, with W the union of all finite initial sign cubes.

Definitions and Normalizations

Let c_0 be the real Banach space of scalar sequences converging to zero, with the supremum norm. The Schur property means that every weakly convergent sequence is norm convergent.

For $n \geq 1$, define

$$W_n = \{(\varepsilon_1, \dots, \varepsilon_n, 0, 0, \dots) : \varepsilon_i \in \{-1, 1\} \text{ for } 1 \leq i \leq n\} \subset c_0$$

and put

$$W = \bigcup_{n=1}^{\infty} W_n.$$

Every element of W belongs to the unit sphere of c_0 .

Proof Intuition

The set W_n is the vertex set of the n -dimensional cube $[-1, 1]^n$, embedded in the first n coordinates of c_0 . Thus the convex hull of W contains every finitely supported point of the unit ball, and these points are dense in B_{c_0} .

The weak-sequential condition holds for a different reason. A weakly convergent sequence in c_0 is coordinatewise convergent. If the sign vectors have unbounded lengths, then after passing to a subsequence every fixed coordinate is eventually a sign, and coordinate convergence forces that coordinate of the weak limit to have modulus one. This would happen for every coordinate, which is impossible for an element of c_0 . So the lengths are bounded, and the sequence lives in a finite set.

Counterexample

Theorem 1. *There is a Banach space X and a subset $W \subset B_X$ such that*

$$\overline{\text{co}}W = B_X,$$

and every weakly convergent sequence of elements of W is norm convergent, but X does not have the Schur property.

Proof. Take $X = c_0$ and $W = \bigcup_{n \geq 1} W_n$ as above.

First we show that $\overline{\text{co}}W = B_{c_0}$. For each fixed n , the convex hull of W_n is exactly the finite cube

$$\{x \in c_0 : \text{supp } x \subseteq \{1, \dots, n\}, |x_i| \leq 1 \text{ for } 1 \leq i \leq n\}.$$

Indeed, this is the usual description of the cube as the convex hull of its 2^n sign vertices. Hence every finitely supported $x \in B_{c_0}$ belongs to $\text{co } W$, after choosing n with $\text{supp } x \subseteq \{1, \dots, n\}$. Since the finitely supported points of B_{c_0} are dense in B_{c_0} by coordinate truncation, $\overline{\text{co}}W = B_{c_0}$.

Now let $(w_j)_j$ be a weakly convergent sequence in W , say

$$w_j \xrightarrow{\text{weak}} x \in c_0.$$

Let n_j be the unique integer such that $w_j \in W_{n_j}$. We claim that the sequence $(n_j)_j$ is bounded. Suppose not. Passing to a subsequence, still denoted $(w_j)_j$, we may assume that $n_j \rightarrow \infty$. For each fixed coordinate k , the coordinate functional $\delta_k \in c_0^* = \ell_1$ gives

$$\delta_k(w_j) \longrightarrow \delta_k(x) = x_k.$$

But for all sufficiently large j , since $k \leq n_j$, we have $\delta_k(w_j) \in \{-1, 1\}$. A convergent scalar sequence whose tail is contained in $\{-1, 1\}$ is eventually constant. Therefore

$$|x_k| = 1$$

for every $k \in \mathbb{N}$. This contradicts $x \in c_0$. Thus the lengths $(n_j)_j$ are bounded.

Choose N such that $n_j \leq N$ for all j . Then

$$\{w_j : j \in \mathbb{N}\} \subseteq \bigcup_{n=1}^N W_n,$$

which is a finite set. Since $(w_j)_j$ is weakly convergent, every subsequence has the same weak limit. A finite-valued convergent sequence in the Hausdorff weak topology must eventually be constant: otherwise two distinct values would occur on infinite subsequences and both would have to equal the weak limit. Hence $(w_j)_j$ is eventually constant, and in particular norm convergent.

Finally, c_0 does not have the Schur property. Let $(e_n)_n$ be the canonical unit vector basis. For every $a = (a_n)_n \in \ell_1 = c_0^*$,

$$a(e_n) = a_n \longrightarrow 0,$$

so $e_n \rightarrow 0$ weakly in c_0 . However, $\|e_n\|_\infty = 1$ for all n , so $(e_n)_n$ is not norm convergent to 0. Therefore c_0 fails the Schur property, while the set W satisfies the two conditions in the source question. $\square$

Verification Notes

The proof is self-contained modulo the standard identification $c_0^* = \ell_1$. The construction also survives harmless variations of the wording in the source question: since $\overline{\text{co}}W = B_X$, any such W must in particular lie inside B_X , and the chosen W already lies in the unit sphere.

The only sequential point is the bounded-length claim. It uses coordinatewise convergence, forced by weak convergence in c_0 , and the fact that a vector whose every coordinate has modulus one cannot belong to c_0 .

Novelty and Duplicate Checks

Before promotion, I searched the run indexes for arXiv 1612.09334, the title terms, **Tsirelson-like spaces**, **Schur property**, **weakly convergent sequence of elements of W** , and related phrases from Remark 5.9(iii). No existing packet for this exact question was found; the nearby hits concerned different Schur, Tsirelson, or Borel complexity questions.

I also ran bounded web searches for exact and near-exact phrases including

- "every weakly convergent sequence of elements of W " "Schur property",
- "Does a Banach space X " "Schur property" "subset W ",
- "Kurka" "subset W " "Schur property",
- " c_0 " "every weakly convergent sequence" "norm convergent" "closed convex hull".

The searches did not surface a prior answer to this embedded question. The packet is therefore marked as a candidate original counterexample pending human review.

References

References

- [1] O. Kurka, *Tsirelson-like spaces and complexity of classes of Banach spaces*, arXiv:1612.09334, 2016.

Human Review Recommendation

This should be reviewed as an elementary counterexample to the displayed question in Remark 5.9(iii). The suggested checks are: verify the transcription of the question, confirm that no additional hidden restriction on W is intended, and inspect the weak-convergence argument for the initial sign-vector set in c_0 . Under the displayed wording, the answer is decisively negative.